Learning Objective 4.4: I can implement a PAM link that transmits and receives ASCII text end-to-end.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Text to bits and back.** The L10 transmitter sends each byte most significant bit first. In ASCII, 'H' is 0x48 and 'i' is 0x69.

   (a) Write the 16 bits the transmitter sends for "Hi", in order.

   (b) A receiver's Repack Bits block is set to LSB first by mistake. What byte does it produce from the first 8 bits, and does it print as a letter?

   (c) A correctly set receiver outputs `01000111 01001111`. What two characters arrived? ('A' is 0x41, and the capital letters follow in order.)

2. **One packet.** A cadet changes the L10 message to `MAYDAY 41.ON 104.8W`, 19 characters. The Vector Source adds a newline, and the packet format is the L10 one: a 64-bit access code, a 32-bit header, then the payload. The link runs at 4 kbaud with one bit per symbol.

   (a) How many bits are in one packet, and how long does it take to send?

   bits: ____

   time: ____

   (b) Write the 16-bit length field, which the header sends twice.

   ____

   (c) What fraction of the bits on the air are text? If the same packets ran at 9.6 kbaud, the rate of L9's channel example, how long would one take?

   text: ____

   time: ____

3. **The access-code search.** Correlate Access Code - Tag Stream compares the last 64 bits with the access code after every new bit and declares a packet when at most 2 bits differ.

   (a) While no packet is arriving, the bits are random. About how many of the 64 match the access code at a typical position, and why?

   (b) The link's bit error rate is 10^{-3}. Find the fraction of packets the receiver misses because of errors in the access code with a threshold of 0, and with a threshold of 2.

   threshold 0: ____

   threshold 2: ____

   (c) On the PlutoSDR version of the link, no packets arrive, yet the constellation after the Costas Loop shows two tight clusters on the real axis. Explain in words what the correlator sees and why, and what the receiver should change.

4. **How much noise a line survives.** In the L10 link, the SNR at the slicer is $8/\sigma^2$ for a noise amplitude σ, each bit is wrong with probability $Q(\sqrt{\text{SNR}})$, and a 31-byte line arrives intact only if all 280 bits of its header and payload are right.

   (a) At $\sigma = 0.9$, find the SNR in dB, the bit error rate, and the fraction of lines that arrive intact. Use $Q(3.14) = 8.4 \times 10^{-4}$.

   SNR: ____

   lines: ____

   (b) The link must deliver 99% of lines. Find the largest bit error rate that allows this, the SNR it needs (use $Q(3.97) = 3.6 \times 10^{-5}$), and the largest σ the link tolerates.

   SNR: ____

   σ: ____

   (c) A teammate proposes 200-byte packets to raise the share of text on the air from 72% to 94%. In words, explain why the link then needs more SNR for the same 99% of packets, and what a Lesson 2 link budget should do about it.

5. **Clock error.** Two radios each derive their clocks from a crystal accurate to 20 ppm, so the receiver's sample clock can differ from the transmitter's symbol clock by up to 40 ppm. The link runs at 4 kbaud.

   (a) A receiver without timing recovery samples at fixed intervals. How many symbols, and how many seconds, until its sampling point slips one whole symbol? How many 344-bit packets is that?

   time: _____

   packets: _____

   (b) The L10 link loses almost no lines up to 10,000 ppm with Symbol Sync in the receiver. Explain why the slip in part (a) does not appear in that link, and why a 40 ppm error is not a concern for it.

   (c) In the clock-error screenshot, the eye sink looks smeared while every line arrives. Explain why the eye sink and Symbol Sync disagree.

Documentation: _____